

Public Economics (ECON 131)

Section #5: Labor Income Taxation

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February 23, 2025

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1 Labor Supply Effects

1.1 Key concepts

- Unlike consumption of regular goods, individuals get disutility from labor. This means that the indifference curves face the other direction (since you prefer less working to more).
- We commonly instead analyze leisure choices (a normal good), allowing us to use similar tools to our analysis across consumption goods.
- We saw before that taxation can affect people's optimal choices of consumption goods. Income taxation similarly affects individuals' optimal labor/leisure decisions by changing the after-tax wage (the return to working an additional hour).
- The effects of taxation can similarly be decomposed into substitution and income effects. We saw before that theoretical analyses of a tax increase is ambiguous since the substitution effect is negative, and income effect is positive. Empirical evidence suggests that the substitution effect tends to be larger than the income effect (on net, taxes decrease labor supply).

1.2 Practice problem: Gruber Ch. 21, combo of Q.1 and Q.7

Suppose that for every hour you work, you can earn \$10 before taxes. Furthermore, suppose for simplicity that you can work up to 10 hours per day, 300 days per year.

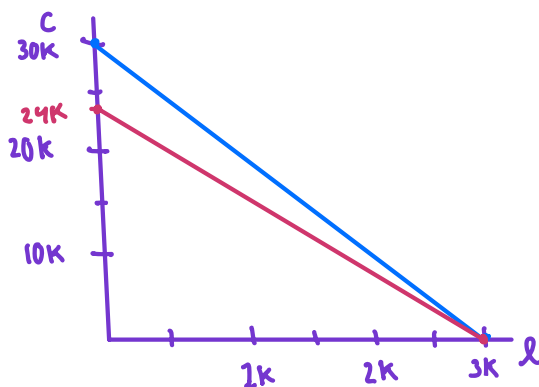
1. Draw the consumption-leisure trade-off under no tax, and under a flat income tax of 20% on all income earned.

w/ no tax
budget constraint?
income = consumption
 $w \cdot h = C$
↑ wage ↑ hours worked
 $10 \cdot [H - l] = C$
↑ how many you can work ↓ leisure

$$10[10 \cdot 300 - l] = C$$

$$10[3000 - l] = C$$

$$30,000 - 10l = C$$



how to draw B.C.?
if $l=0$, $C=30,000$
if $C=0$, $10l=30,000$
 $l=3000$

$$\text{Slope} = -\frac{30,000}{3,000} = -10$$

w/ tax $\downarrow$ no longer keeping 100%
 $t = 0.2$
 $h \cdot w \cdot (1-t) = C$
 $[3000 - l] \cdot 10(0.8) = C$
 $[3000 - l] 8 = C$
 $24,000 - 8l = C$

how to draw B.C.?
if $l=0$, $C=24,000$
if $C=0$, $24,000 = 8l$
 $l=3000$

$$\text{slope} = -\frac{24,000}{3,000} = -8$$

2. Under the above tax regimes, draw a set of representative indifference curves such that (1) the income effect of the tax dominates the substitution effect; (2) the substitution effect of the tax dominates the income effect.

first, let's think about it conceptually!

	C	l	h
substitution	↓	↑	↓
income effect	↓	↓	↑
total	↓	?	?

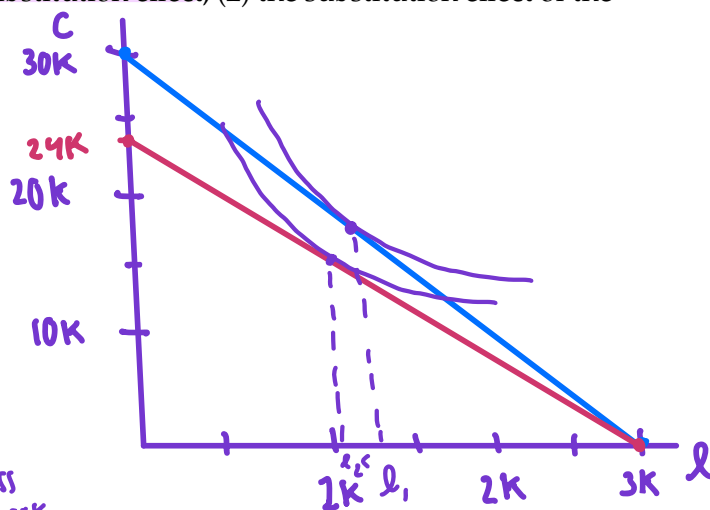
feel richer or poorer?

consume less of both C & l → consume less leisure, work more

which is costlier? C or l?

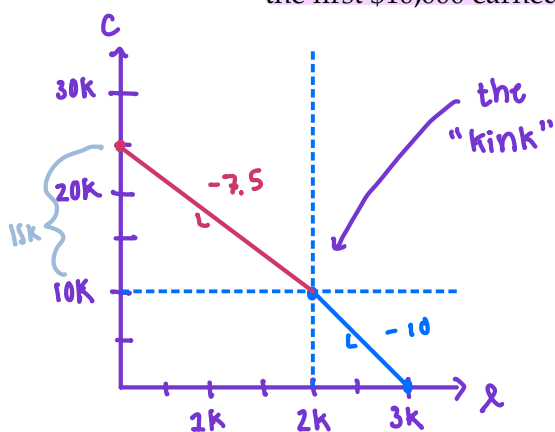
consume less of C and more of l

before, one hour of leisure costed you \$10 of income, after tax, it only costs you \$8.



for the income effect to dominate, the total effect of l should be ↓ or ↑?
 $l_1 > l_2$

3. Now draw the consumption-leisure trade off under an income tax where you pay no tax on the first \$10,000 earned and a tax of 25% on all income over \$10,000.



where is the "kink" [where does the budget constraint change]

@ \$10,000 and below $t = 0$

→ how many hours do you need to work to make \$10k?

$$30,000 - 10l = C = 10,000$$

$$20,000 = 10l$$

$$l = 2,000$$

@ above \$10,000, $t = 0.25$

vertical intercept

calculate total income earned for working all 3k hours

$$1,000(10) + 2,000 \cdot 10 \cdot (1 - 0.25) = 10,000 + 20,000(0.75)$$

$$\Rightarrow 10,000 + 15,000 = 25,000$$

$$\text{slope} = -10(1 - 0.25) = -7.5$$

4. Now draw the consumption-leisure trade off under an income tax where you pay 10% on the first \$5,000 earned, 20% on the next \$10,000 earned, and 30% thereafter.

as above, if $l = 3,000 \Rightarrow C = 0$

now, the first kink

earn 5,000 and $t = 0.10$,

how much do you get to keep?

$$5,000(1 - 0.10) = 5,000(0.9) = 4,500!$$

$$4,500 = w \cdot (1 - t) \cdot h$$

$$4,500 = 10(1 - 0.10)[3000 - l]$$

$$4,500 = 9[3000 - l]$$

$$4,500 = 27,000 - 9l$$

$$9l = 22,500$$

$$l = 2,500$$

$$\text{slope} = -10(1 - 0.10)$$

$$= -10(0.9)$$

$$= -9$$

now, the second kink

earn 10,000 and $t = 0.20$

how much do you get to keep?

$$10,000(1 - 0.2) = 10,000(0.8) = 8,000$$

so, now you have made

$$4,500 + 8,000 = 12,500$$

where's the kink? missing l

$$8,000 = 10 \cdot (1 - 0.2)[2500 - l]$$

$$8,000 = 8[2500 - l]$$

$$1000 = 2500 - l$$

$$l = 1500$$

$$\text{slope} = -10(1 - 0.2)$$

$$= -10(0.8)$$

$$= -8$$

vertical intercept

30% tax after earning 15,000

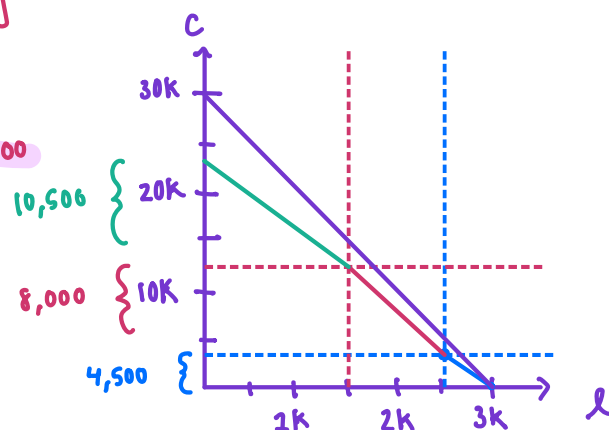
$$C = 12,500 + 10(0.7) \cdot [1500]$$

$$C = 12,500 + 7[1500]$$

$$C = 12,500 + 10,500$$

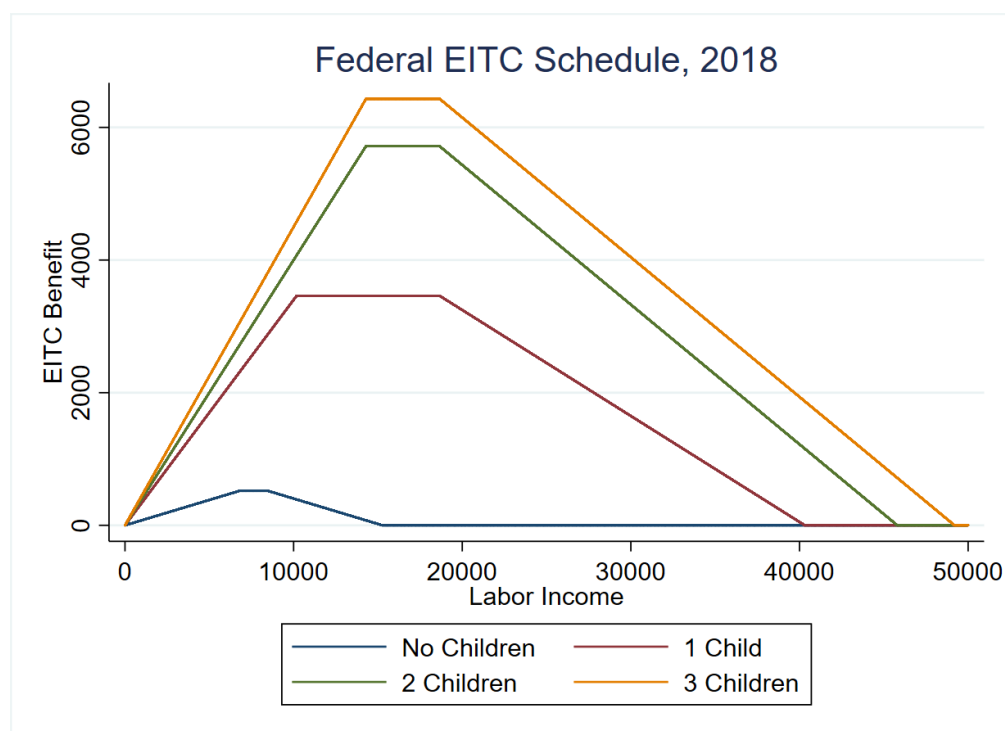
$$C = 23,000$$

$$\text{slope} = -7$$



1.3 Earned Income Tax Credit

One way to make the tax system progressive is by increasing the marginal tax rate as income goes up, so top earners pay higher tax rates than low earners. Another way to make the tax system progressive is by introducing tax credits and transfers to low earners: this can make their post-tax income higher than their pre-tax income. One of the main transfers in the US is the Earned Income Tax Credit (EITC), which is administered as a refundable income tax credit (if you do not owe any taxes, you get a tax refund). The size of the EITC benefit depends prominently on marital status, number of children and household labor income (graphed below for singles): In contrast

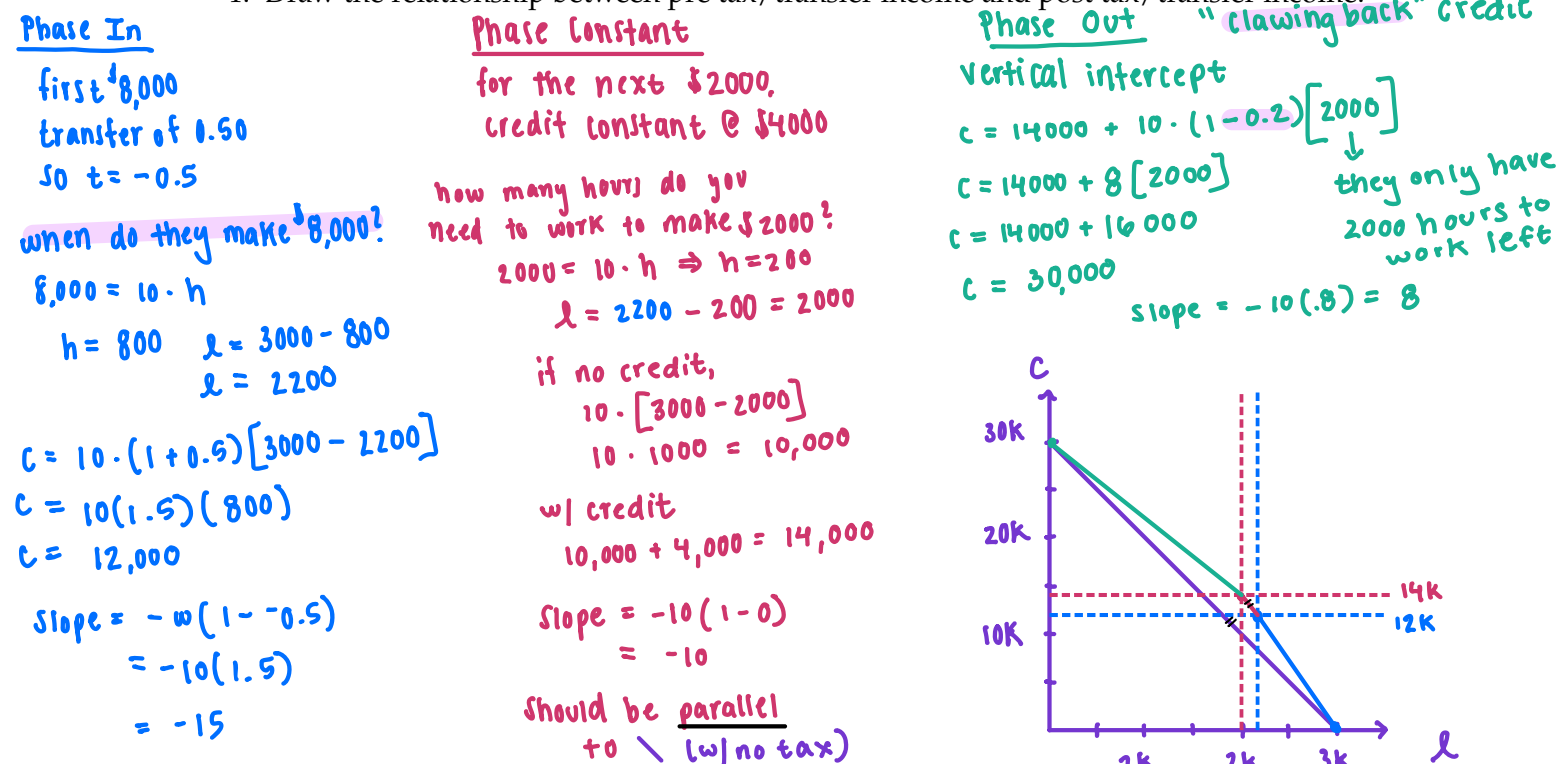


to other means-tested safety-net programs, you are only eligible for the EITC if you work (you can see this above). As we will see in the next problem, the EITC has this three-slope structure in part to incentivize labor supply.

1.4 Practice Problem: Gruber, Ch.21, Q.8

Starting from the previous problem's no-tax setup, suppose that the government introduces an Earned Income Tax Credit such that for the first \$8,000 in earnings, the government pays 50 cents per dollar on wages earned. For the next \$2,000 of earnings, the credit is held constant at \$4,000, and after that point the credit is reduced at a rate of 20 cents per dollar earned.

1. Draw the relationship between pre tax/transfer income and post tax/transfer income.



2. Mark on your graph the portions of the income schedule where the labor supply effects of the policy are positive, negative, or ambiguous, relative to no tax.

yes credit no credit

$$-15 < -10$$

→ 2 hr of leisure
reduces consumption
by \$10 (\$15)
it has gotten
costlier!
consume less

$$-8 > -10$$

leisure has
gotten cheaper

	Phase In		Phase Constant		Phase Out	
	l	h	l	h	l	h
income effect	↑	↓	↑	↓	↑	↓
substitution effect	↓	↑	NONE 0	NONE 0	↑	↓
total/net effect	?	?	↑	↓	↑	↓

comparisons

→ income effect: levels

→ substitution effect: slopes

2 Optimal Labor Income Taxation

2.1 Key Concepts

Laffer Curve

- The **Laffer Curve** illustrates the relationship between tax rates and government revenue
- Tax revenue = tax rate $\times$ tax base
- Increasing a tax has two opposing effects on the revenue:
 - Mechanical: higher rates mean higher revenue per dollar of taxable income
 - Behavioral: higher rates may cause people to work less, decreasing the tax base
- At low tax rates, tax rate increases likely increase revenue on net. But if tax rates are too high, then tax base erosion may be higher than rate increases (leading to decreased revenue). There's thus a theoretical turning point at which revenue is maximized.
- Empirical evidence suggests that the US (and most other countries) is currently operating on the upward sloping portion of the curve.

Picking an optimal tax

- The definition of the optimal labor income tax will depend on your social welfare criteria
- If the social planner is Rawlsian, wants to maximize the utility of the worst-off person in society $\rightarrow$ maximize tax revenue (ie find the τ^* at the peak of the Laffer curve)

$$\tau = \frac{1}{1 + e}$$

- If the social planner is utilitarian, the optimal tax formula will capture the equity-efficiency trade-off:

$$\tau = \frac{1 - \bar{g}}{1 - \bar{g} + e} \text{ where } \bar{g} = \frac{\sum_i z_i \cdot \frac{\partial u^i}{\partial c}}{Z \cdot \sum_i \frac{\partial u^i}{\partial c}}$$

where $\bar{g}$ can be thought of as a measure of equality in society (1 = perfect equality, 0 = perfect inequality), and e is the labor supply elasticity with respect to taxes (CBO: $e \approx 0.4$).